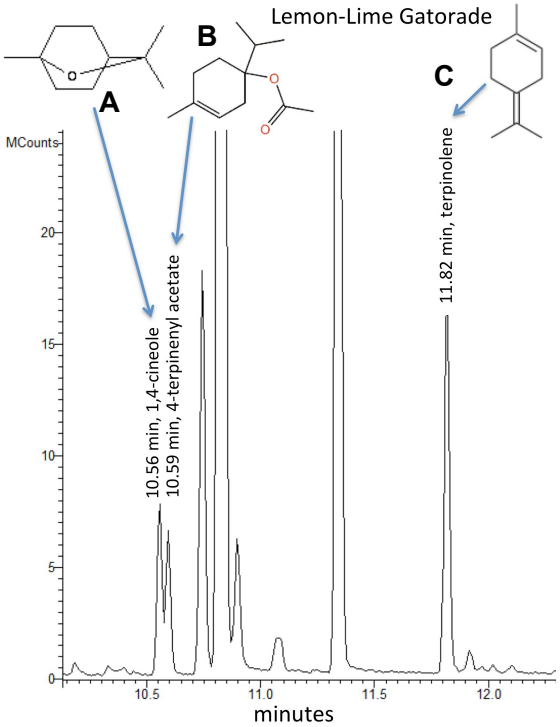
(23 points) On the right is a portion of a total ion chromatogram obtained from GC-MS of a sample of the headspace that was taken over Lemon-Lime Gatorade sports drink.

The length of the column is **30.0 m** and flow rate is **1.00 L/min**.

Potentially useful equations are on the final page.

Information about the mobile phase and three of the peaks is summarized below:



Chemical Species	Simple label to refer to these compounds	Retention time, min	Peak width at base, min	Molecular mass, g/mol
Mobile phase, helium	--	1.47	--	--
1,4-cineole	A	10.56	0.050	154.25
4-terpinenyl acetate	B	10.59	0.050	196.29
terpinolene	C	11.82	0.060	136.23

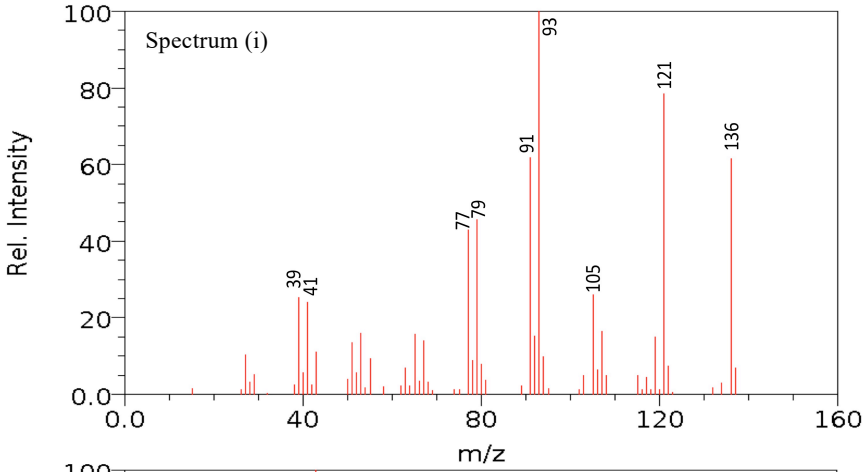
(a) (4 points) Which solute, A, B, or C, has the largest partition coefficient (distribution constant)? (Identify the compound and explain reasons for answer. Calculations are not needed here).

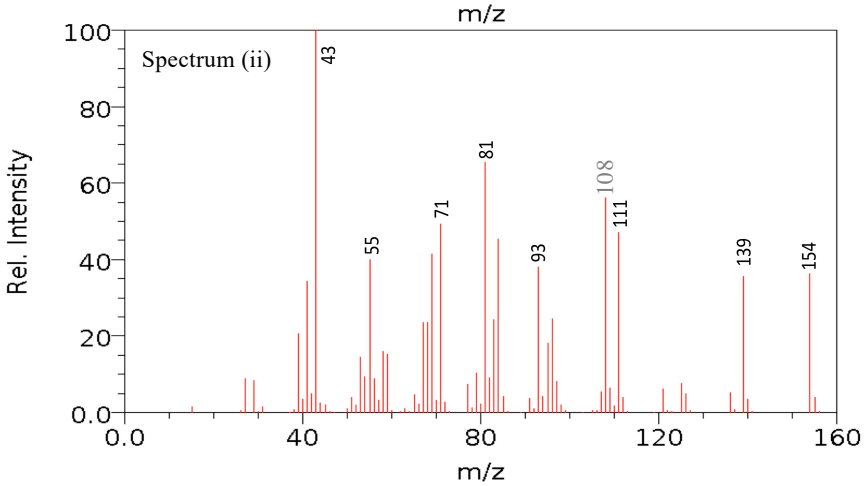
(b) (4 points) Calculate the number of plates for the column based on the retention time and peak width for (B) 4-terpinenyl acetate.

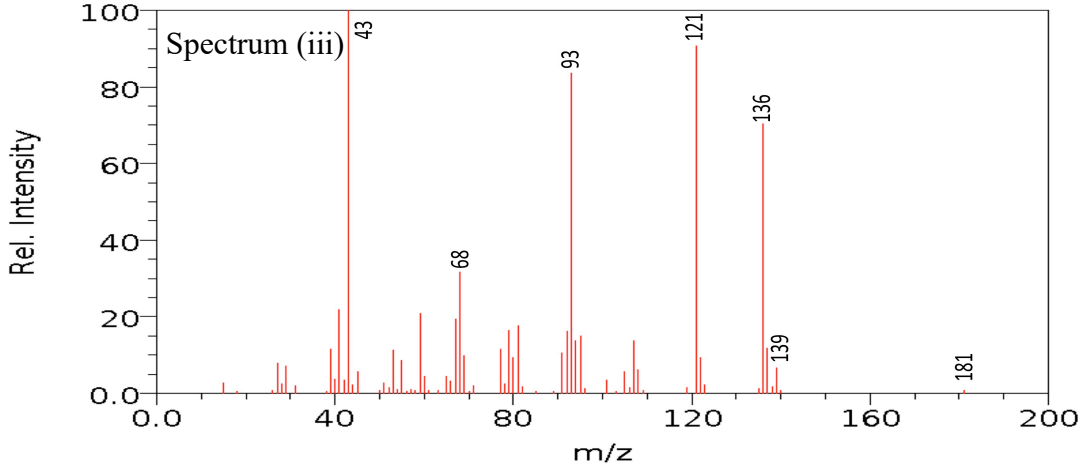
(c) (4 points) What is the plate height for the column, given the number of plates determined in part (b)?

(d) (4 points) Calculate the resolution for (B) 4-terpinenyl acetate and (C) terpinolene.

(e) (3 points) Match each of the species, **(A) 1,4-cineole**, **(B) 4-terpinenyl acetate**, and **(C) terpinolene** to its electron ionization-mass spectrum number (i, ii, iii) on the following page. ***Refer to table within this problem for molecular mass.***







(f) (4 points) In the **topmost mass spectrum** in part (e) above, select **one peak** can be described by the following terms. Note that some of these terms have multiple answers. Also, some peaks can be described by several of these terms.

Select peaks from the following: m/z 39, 41, 77, 79, 91, 93, 105, 121, 136.

radical cation_____

molecular ion_____

parent ion_____

base peak_____

$$K_A = \frac{C_{s,A}}{C_{m,A}}$$

$$k'_A = \frac{moles_s}{moles_m} = \frac{C_s V_s}{C_m V_m} = K_A \frac{V_s}{V_m} = \frac{(t_{r,A} - t_m)}{t_m}$$

$$\alpha = \frac{t_{r,B} - t_m}{t_{r,A} - t_m} = \frac{t'_{r,B}}{t'_{r,A}} = \frac{k'_B}{k'_A} = \frac{K_B}{K_A} \quad \text{where } t'_{r,B} > t'_{r,A}$$

$$R_s = \frac{2\,(t_{r,B} - t_{r,A})}{w_A + w_B} = \frac{\Delta t_r}{w_{av}}$$

$$\sigma^2 = H\,x$$

$$\sigma^2 = H\,L$$

$$N = \frac{L}{H}$$

$$N = \frac{L}{H} = \frac{L^2}{\sigma^2} = \frac{16\,L^2}{w^2}$$

$$N = \frac{16\,t_r^2}{w^2} = \frac{5.55\,t_r^2}{w_{1/2}^2}$$